

Surname

Name

University ID Number

Signature

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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 4. Additional papers will not be considered.
 - Clarity, order and precision are strongly considered for the final evaluation.
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1. [Kalman filter]

Consider the following dynamic system

$$\begin{aligned}x_1(t+1) &= \frac{1}{2}x_2(t) + v_1(t) \\x_2(t+1) &= \frac{1}{2}x_2(t) + v_2(t) \\y(t) &= x_2(t)\end{aligned}$$

where $v_1 \sim WN(0,1)$, $v_2 \sim WN(0,1)$ and $v_1(t) \perp v_2(t)$. Answer the following questions.

- Is the steady-state 1-step ahead state prediction error bounded?
- Compute the steady-state 1-step ahead prediction error variance of x_2 and comment the result.
- Write the equations of the steady-state Kalman filter for the process x_2 .
- Compute the steady-state state estimation error variance of x_2 and comment the obtained result.

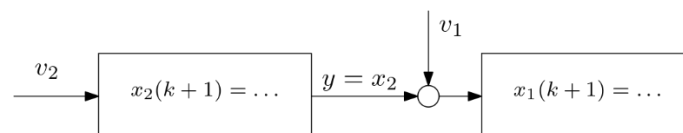
Solutions

a. The system matrices are:

$$F = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/2 \end{bmatrix}, H = \begin{bmatrix} 0 & 1 \end{bmatrix}, \Gamma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, V_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, V_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, V_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the first asymptotic convergence theorem, since the system is asymptotically stable and V_{12} is null, the 1-step ahead state prediction error is indeed bounded.

- b. Inspecting the state equations, it can be easily seen how the system is the series connection of two subsystems. The following block diagram clarifies the connection.



In particular, the process x_2 does not depend on x_1 and can be thus studied independently from x_1 :

$$\begin{aligned}x_2(t+1) &= \frac{1}{2}x_2(t) + v_2(t) \quad \rightarrow \quad F = [1/2], H = [1], V_1 = [1], V_2 = [0], V_{12} = [0] \\y(t) &= x_2(t)\end{aligned}$$

The corresponding Riccati equation (DRE) is

$$P(t+1) = \frac{1}{4}P(t) + 1 - \frac{1/4 P(t)^2}{P(t)} = 1$$

The DRE and the ARE coincide. Thus the 1-step ahead prediction error variance of x_2 is

$$P(t+1) = P(t) = \bar{P} = 1$$

The 1-step ahead prediction error variance is equal to the process noise variance v_2 . In fact, as the output equation is not affected by noise, in predicting the future state the only source of uncertainty lies in the process noise.

- c. The equations of the Kalman filter can be easily derived from the Kalman predictor ones.

The Kalman predictor equations are:

$$\begin{cases} \hat{x}_2(t+1|t) = \frac{1}{2}y(t) \\ \hat{y}(t|t-1) = \hat{x}_2(t|t-1) \end{cases}$$

The Kalman filter equations can be derived with a 1-step backward state projection since the matrix F is invertible.

$$\begin{cases} \hat{x}_2(t|t) = F^{-1}\hat{x}_2(t+1|t) = 2 \cdot \frac{1}{2}y(t) = y(t) \\ \hat{y}(t|t) = \hat{x}_2(t|t) \end{cases}$$

- d. The steady-state estimation error variance $Var[\epsilon(t|t)]$ can be computed as follows:

$$P = Var[\hat{x}_2(t+1|t) - x_2(t+1|t)] = Var\left[F \cdot \hat{x}_2(t|t) - \frac{1}{2}x_2(t) - v_2(t)\right] = Var\left[\frac{1}{2}\epsilon(t|t) - v_2(t)\right]$$

Since the estimation error is uncorrelated with the process noise, the previous expression becomes:

$$P = Var\left[\frac{1}{2}\epsilon(t|t) - v_2(t)\right] = Var\left[\frac{1}{2}\epsilon(t|t)\right] + Var[v_2(t)],$$

which eventually gives the result:

$$1 = \frac{1}{4}Var[\epsilon(t|t)] + 1 \Rightarrow Var[\epsilon(t|t)] = 0$$

The result does not surprise. In fact, since no noise affects the measure of $x_2(t)$, its knowledge at time “t” is always perfect (i.e., the estimation variance is null). This also explains why the Kalman filter state equation trivially states that the state estimate coincides with the measured output.

[Stochastic processes/Prediction]

Consider the processes, described by the following equations

$$\begin{aligned} z(t) &= \frac{1}{2}z(t-1) + e(t) + 2e(t-1) \\ y(t) &= z(t-1) + \eta(t) \end{aligned}$$

where $e \sim WN(0,1)$, $\eta \sim WN(0, \lambda_\eta^2)$ and $e(t) \perp \eta(t)$ and $\lambda_\eta^2 \geq 0$. Answer the following questions:

- Discuss if $z(t)$ and $y(t)$ are stationary processes.
- Compute the power spectral density of $y(t)$ for a generic value of λ_η^2
- For $\lambda_\eta^2 = 0$, compute the 1-step ahead predictor of $y(t)$ (based only on past measures of $y(t)$).
- For $\lambda_\eta^2 = 0$, compute the 1-step ahead prediction error variance of $y(t)$.
- For $\lambda_\eta^2 = 1$, compute the 1-step ahead predictor of $y(t)$ (based only on past measures of $y(t)$).

Solutions

- The process $z(t)$ is stationary, being the output of a stable filter with transfer function $G(z)$

$$G(z) = \frac{1 + 2z^{-1}}{1 - \frac{1}{2}z^{-1}},$$

fed by a stationary white noise $e(t)$. Also, the process $y(t)$ is stationary, being the sum of two stationary processes $z(t-1)$ (the delay does not affect the stationarity of the process $z(t)$) and $\eta(t)$.

- The process $z(t)$ is uncorrelated with the process $\eta(t)$, as they originate from uncorrelated noises. Thus, the power spectral density (PSD) of $y(t)$ can be computed as:

$$\Gamma_y = \Gamma_z + \Gamma_\eta$$

Notice that, being $z(t)$ a stationary process, the 1-step delay $z(t-1)$ has no effect on the covariance function. The power spectral density $\Gamma_\eta = \lambda_\eta^2$, since $\eta(t)$ is a white noise. The PSD of Γ_z can be computed as follows:

$$\Gamma_z = |G(e^{j\omega})|^2 \lambda_e^2 = \left| \frac{1 + 2e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}} \right|^2 \cdot 1 = \frac{5 + 4 \cos \omega}{\frac{5}{4} - \cos \omega}$$

Eventually:

$$\Gamma_y = \frac{5 + 4 \cos \omega}{\frac{5}{4} - \cos \omega} + \lambda_\eta^2.$$

- When $\lambda_\eta^2 = 0$, the noise $\eta(t)$ does not contribute to the process $y(t)$, which can be written as:

$$y(t) = G(z) \cdot z^{-1} \cdot e(t) = \frac{z^{-1} + 2z^{-2}}{1 + \frac{1}{2}z^{-1}} \cdot e(t)$$

Firstly, the process is written in canonical form, yielding:

$$y(t) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \cdot \delta(t)$$

where $\delta(t) \sim WN(0,4)$. The 1-step ahead predictor is hence:

$$y(t+1|t) = \frac{1}{1 + \frac{1}{2}z^{-1}} \cdot y(t)$$

- d. The prediction error variance, for the 1-step ahead prediction is equal to the variance of the noise that feeds the stochastic process, in this case $\delta(t)$. Thus:

$$Var[\epsilon(t+1|t)] = Var[\delta] = 4$$

- e. When $\lambda_\eta^2 = 1$, the effect of η on the measured variable is no longer zero, and an equivalent stochastic description for $y(t)$ must be found. To find such description, the PSD previously found is leveraged.

$$\Gamma_y = \frac{5 + 4 \cos \omega}{\frac{5}{4} - \cos \omega} \cdot 1 + \lambda_\eta^2 = \frac{5 + 4 \cos \omega}{\frac{5}{4} - \cos \omega} + 1 = \frac{(5 + 4 \cos \omega) + \frac{5}{4} - \cos \omega}{\frac{5}{4} - \cos \omega} = \frac{\frac{25}{4} + 3 \cos \omega}{\frac{5}{4} - \cos \omega}$$

Recalling that the PSD of generic ARMA(1,1) is:

$$\Gamma = \frac{(1 + c^2) + 2c \cos \omega}{(1 + a^2) + 2a \cos \omega} \cdot \lambda^2$$

The following coefficients are found, comparing the two expressions:

$$a = -\frac{1}{2} \quad c \approx 0.25 = \frac{1}{4} \quad \lambda^2 = 6$$

(when computing the coefficients, the solution $c \approx 4$, despite being correct, is immediately discarded, as it would lead to an equivalent ARMA model not in canonical form). The stochastically equivalent model for $y(t)$ is thus:

$$y(t) = \frac{1 + \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1}} \cdot \sigma(t)$$

where $\sigma(t) \sim WN(0,6)$. The 1-step ahead predictor is then:

$$y(t+1|t) = \frac{\frac{3}{4}}{1 + \frac{1}{4}z^{-1}} \cdot y(t)$$

[Grey-box system identification]

Explain and discuss how the Kalman Filter can be used to solve grey-box system-identification problems

[See lecture notes](#)

[state-space identification]

Explain and discuss the “realization” of a transfer function into a state-space representation

[See lecture notes](#)